PROGRAM:

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#include<stdio.h>
#define MAX_SIZE 100
void createArray(int arr[], int *size);
void insertElement(int arr[], int *size, int element, int position);
int searchElement(int arr[], int size, int element);
void deleteElement(int arr[], int *size, int position);
void displayArray(int arr[], int size);
int main() {
        int arr[MAX_SIZE];
        int size = 0;
        int choice, element, position, result;
        while(1) {
                printf("\nArray Operation Menu:\n");
                printf("1. Create Array\n");
                printf("2. Insert Element\n");
                printf("3. Search Element\n");
                printf("4. Delete Element\n");
                printf("5. Display Array\n");
                printf("6. Exit\n");
                printf("Enter your choice: ");
                scanf("%d", &choice);
                switch(choice) {
                        case 1:
                                createArray(arr, &size);
                                break;
                        case 2:
                                printf("Enter element to insert: ");
                                scanf("%d", &element);
                                printf("Enter position to insert (0 to %d): ", size);
                                scanf("%d", &position);
                                insertElement(arr, &size, element, position);
                                break;
                        case 3:
                                printf("Enter element to search: ");
                                scanf("%d", &element);
                                result = searchElement(arr, size, element);
                                if(result != -1)
                                        printf("Element fount at position: %d\n", result);
                                else
                                        printf("Element not found in the array\n");
                                break;
                        case 4:
                                printf("Enter position to delete (0 to %d): ", size-1);
                                scanf("%d", &position);
                                deleteElement(arr, &size, position);
                                break;
                        case 5:
                                displayArray(arr, size);
                                break;
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                case 6:
                        return 0;
                default:
                        printf("Invalid choice! Please enter a valid option.\n");
                }
        }
        return 0;
}
void createArray(int arr[], int *size) {
        int n, i;
        printf("Enter the number of elements: ");
        scanf("%d", &n);
        if(n> MAX_SIZE) {
                printf("Error: Number of elements exceeds the maximum array size.\n");
                return;
        }
        printf("Enter %d elements: ",n);
        for(i=0; i <n; i++) {
                scanf("%d", &arr[i]);
        }
        *size = n;
        printf("Array created successfully.\n");
}
void insertElement(int arr[], int *size, int element, int position) {
        int i;
        if(*size >= MAX_SIZE) {
                pirntf("Error: Array is full. Cannot insert element.\n");
                return;
        }
        if((position < 0) || (position > *size)) {
                printf("Error: Invalid position.\n");
                return;
        }
        for(i=0; i < size; i++) {
                arr[i] = arr[i – 1];
        }
        arr[position] = element;
        (*size)++;
        printf("Element inserted successfully.\n");
}
int searchElement(int arr[], int *size, int position) {
        int i;
        for(i = 0; i < size; i++) {
                if(arr[i] == element) {
                        return i;
                }
        }
        return -1;
}
void deleteElement(int arr[], int *size, int position) {
        int i;
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        if((position < 0) || (position >= *size)) {
                printf("Error: Invalid position.\n");
                return;
        }
        for(i = position; i < *size - 1; i++) {
                arr[i] = arr[i + 1];
        }
        (*size) --;
        printf("Element deleted successfully.\n");
}
void displayArray(int arr[], int size) {
        int i;
        if(size == 0) {
                printf("Array is empty.\n");
                return;
        }
        printf("Array elements: ");
        for(i =0; i < size; i++) {
                printf("%d", arr[i]);
        }
        printf("\n");
}
```